Nokia Core Node Failure SOP: QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) on BLR-CORE-02

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Critical | **Alarm Code:** ALARM_6257

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution procedures for a QoS congestion event (microbursts, interface tail drops, QoS queue starvation) on the Nokia Core node BLR-CORE-02. The Nokia Core node, a critical component of the network infrastructure, is responsible for routing and forwarding traffic between different network segments. The QoS congestion event occurs when the node's QoS mechanisms are unable to adequately manage traffic congestion, leading to microbursts, interface tail drops, and QoS queue starvation. This can be attributed to various factors, including misconfigured QoS policies, inadequate QoS monitoring, and hardware limitations. The underlying protocol mechanics of the QoS mechanism are based on the concept of hierarchical QoS, where traffic is classified and policed based on its priority and rate. The hardware implementation of the QoS mechanism utilizes a combination of software and firmware to allocate resources and manage traffic flow. The diagnostic and resolution procedures outlined in this SOP are designed to identify and address the root cause of the QoS congestion event, ensuring the optimal performance and reliability of the Nokia Core node.

Pre-requisite Checks

- item: Verify the site location of the Unknown Site hosting the Nokia Core node BLR-CORE-02., description: Ensure the site location is accurately identified to facilitate remote hands dispatch and access to the node.

- item: Check the status of redundant links connected to the Nokia Core node BLR-CORE-02., description: Verify that redundant links are operational and ready to assume traffic load in case of a failure.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate QoS monitoring	Medium
Hardware limitations	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy all, Expected Output: Display all QoS policies configured on the node, Description: This command retrieves a list of all QoS policies configured on the node, including policy
```

```
names, priorities, and rates. This information can be used to identify potential misconfigurations or anomalies.

Command: show qos monitor all, Expected Output: Display all QoS monitors configured on the node, Description: This command retrieves a list of all QoS monitors configured on the node, including monitor names and settings. This information can be used to identify potential inadequacies in QoS monitoring.

Command: show interface all, Expected Output: Display interface statistics and configuration, Description: This command retrieves interface statistics and configuration, including error counts and packet drops. This information can be used to identify potential hardware limitations or issues.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configurations, Command: configure qos policy priority rate , Description: This command corrects misconfigured QoS policies by verifying and updating policy priorities and rates. Ensure that QoS policies are properly configured to manage traffic flow and prevent congestion events.

Step 2: Step: Implement QoS monitoring enhancements, Command: configure qos monitor threshold , Description: This command enhances QoS monitoring by implementing threshold settings to detect and alert on potential congestion events. Ensure that QoS monitors are properly configured to detect anomalies and trigger corrective actions.

Step 3: Step: Upgrade or replace hardware components, Command: upgrade hw-component revision , Description: This command upgrades or replaces hardware components to address potential hardware limitations or issues. Ensure that hardware components are properly upgraded or replaced to prevent congestion events.

Post-Check Verification

```
Command: show qos policy all, Expected Output: Display all QoS policies configured on the node, Description: This command verifies that QoS policies are properly configured to manage traffic flow and prevent congestion events.

Command: show qos monitor all, Expected Output: Display all QoS monitors configured on the node, Description: This command verifies that QoS monitors are properly configured to detect anomalies and trigger corrective actions.

Command: show interface all, Expected Output: Display interface statistics and configuration, Description: This command verifies that interface statistics and configuration are within normal ranges and do not indicate potential hardware limitations or issues.
```

Roll-back Procedures

- Step: Restore original QoS policy configurations, Command: configure qos policy priority rate , Description: This command restores original QoS policy configurations in case of a rollback. Ensure that QoS policies are properly restored to prevent congestion events.

- Step: Disable QoS monitoring enhancements, Command: configure qos monitor threshold , Description: This command disables QoS monitoring enhancements in case of a rollback. Ensure that QoS monitors are properly disabled to prevent incorrect detection of congestion events.

- Step: Downgrade or replace hardware components, Command: downgrade hw-component revision , Description: This command downgrades or replaces hardware components in case of a rollback. Ensure that hardware components are properly downgraded or replaced to prevent congestion events.